Statistical validation of TESS and Kepler candidates

TRICERATOPS validation pipeline — Observer's Guide
with the graphical user interface

Pipeline **triceratops_validation.py v4.2**
GUI **triceratops_gui.py v1.2**
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Abstract

This guide is written for observers and collaborators who want to use the validation pipeline without reading its full technical manual. Part I explains what TRICERATOPS computes, which false-positive scenarios it considers, what the pipeline adds around it, and how ground-based photometry enters the analysis. Part II explains the graphical user interface (GUI), installation in a general working directory, profiles, and a worked example (AAVSO campaign 942 on TIC 65344135). Part III is a complete reference of all options (including Kepler mode introduced in v4.2), output files and common problems. The technical manual remains the reference for methods and tests.

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Part I

Background

1. Why statistical validation?

A transit candidate is a periodic dip in a light curve. A planet crossing its star produces such a dip, but so do several other configurations: an eclipsing binary (EB) on the target (possibly grazing or diluted); an EB in the background whose light is mixed with the target (blend); a planet or EB around a fainter bound companion; or an eclipsing binary on a nearby star inside the same photometric aperture (nearby eclipsing binary, NEB).

For many TESS and Kepler candidates—especially faint stars and long periods—the planet's mass cannot be measured with radial velocities. Statistical validation then asks: given everything we know about the star, its surroundings and the shape of the light curve, how probable is it that the signal is a planet on the target?

What this pipeline does. `triceratops_validation.py` prepares the photometry (TESS FFI cutouts or Kepler long-cadence data), checks the candidate with diagnostic tests, and calls the TRICERATOPS package (Giacalone et al. 2021) to compute false-positive probabilities. From version 4.1 it can use ground-based light curves to rule out nearby stars and to test colour dependence of the depth. From version 4.2 it also supports Kepler targets. The GUI `triceratops_gui.py` lets you set all of this up in a window, run it and inspect the results.

2. How TRICERATOPS works

2.1 Scenarios

TRICERATOPS compares the observed transit with astrophysical scenarios. Only TP, PTP and DTP place a planet on the target; all other scenarios are false positives.

Scenario	Meaning
TP	Transiting planet on the target star.
EB, EBx2P	Target is an eclipsing binary (or at twice the period).
PTP, PEB, …	Unresolved bound companion; planet/EB on the primary.
STP, SEB, …	Planet/EB on the unresolved bound secondary.
DTP, DEB, …	Planet/EB on the target, diluted by a chance-aligned star.
BTP, BEB, …	Planet/EB on an unresolved background star.
NTP, NEB, …	Planet/EB on a nearby resolved star in the aperture.

2.2 Probabilities

For each scenario S, TRICERATOPS computes a probability proportional to the prior p(S) times the integral of the likelihood over the unknown parameters (Monte Carlo, typically ~10^6 draws). Priors use planet occurrence rates, companion statistics, and a Galactic model (TRILEGAL) for background stars. The summary numbers are:

FPP = 1 – (P_{TP} + P_{PTP} + P_{DTP}) — probability the signal is *not* a planet on the target.
NFPP = sum of nearby-star scenarios (NTP, NEB, …).

2.3 When is a candidate "validated"?

Following Giacalone et al. (2021), the pipeline reports **PASSED** when FPP < 0.015 and NFPP < 0.001. Other combinations: FPP < 0.5 with NFPP < 0.001 → likely planet; NFPP > 0.1 → likely nearby false positive. A PASSED is a statistical statement, meaningful only when diagnostics show nothing suspicious, and is no substitute for follow-up.

2.4 What lowers the false-positive probability

- **Clean transit shape** — U-shaped, correct duration for the star.
- **Precise depth and correct dilution** — wrong dilution → wrong radius.

- **High-resolution imaging** — contrast curve limits close companions.
- **Ground-based photometry** — resolves which star dims; cleared stars remove NEB/NTP scenarios.

2.5 Limitations

TRICERATOPS does not replace spectroscopy or high-resolution imaging when those are available. It assumes the input ephemeris and dilution model are reasonable; the pipeline's diagnostic tests (odd/even, secondary search, difference imaging, ETV, ground clearance) exist to catch cases where that assumption fails. Very sparse long-period light curves can leave period aliases that only ground timing resolves.

3. What the pipeline does around TRICERATOPS

Before calling TRICERATOPS the pipeline: builds multi-sector (TESS) or multi-quarter (Kepler) light curves; detrends with transit-aware windows; searches or refines the ephemeris; measures per-transit depths and mid-times; runs difference-image centroids; optional limb-darkened fit; secondary-eclipse search; ETV analysis; and optional neighbour vetting. After TRICERATOPS it writes a SUMMARY block, `summary_<id>.json`, plots and tables. Optional ground light curves clear stars and, with TRICERATOPS+, enter the multi-colour likelihood.

4. Ground-based observations in the analysis

4.1 The blending problem

TESS pixels are large (~21″). A dip attributed to the target may originate on a neighbour. Ground photometry with a smaller aperture can show which star dims and clear the others.

4.2 What the pipeline does with a ground light curve

It fits a transit (free depth and mid-time), computes the expected depth for every nearby star given the aperture, seeing and band magnitudes, and classifies stars as consistent, excluded or allowed. Excluded stars are removed before calc_probs (unless `--ground-no-clear`). Chromaticity across bands is reported. With TRICERATOPS+ up to seven band light curves can enter calc_probs (`--tri-plus`).

4.3 Practical advice for observers

- **Timing.** Cover the whole transit with ≥45–60 min baseline; add ephemeris uncertainty. Use the GUI tool *Predicted transit times*.
- **Band.** Red (Rc / Sloan r′) for S/N; a second blue band enables the colour test.
- **Apertures.** Measure target and each relevant neighbour separately; report radius and FWHM in arcseconds.
- **Deliver raw relative fluxes** with airmass if possible—not a pre-detrended curve.
- **Times.** Prefer BJD_TDB; JD_UTC / HJD_UTC are accepted if labelled.
- **Formats.** AstroImageJ tables, AAVSO/EXOTIC reports, or simple time/flux/error tables.

Part II

Using the GUI

5. Installation and start

5.1 What you need

The scripts. Place `triceratops_validation.py`, `triceratops_gui.py`, and the folders `profiles/` and `tests/` together in one working directory—the folder from which you run the GUI or the pipeline. Any path on Linux, macOS, Windows or WSL is fine.

A Python environment for the pipeline. Needs triceratops ≥ 1.1, lightkurve, astroquery, wotan, transitleastsquares, numpy, scipy, matplotlib, astropy. Optionally a second environment `triplus` for TRICERATOPS+:

```
conda create -n triplus -c conda-forge python=3.11 "numpy<2.4" scipy pandas matplotlib \
astropy astroquery numba celerite emcee seaborn arviz corner pytest
conda activate triplus
pip install lightkurve "pytransit==2.2.0" mechanicalsoup pyrr wotan \
transitleastsquares "setuptools<81"
git clone https://github.com/JGB276/TRICERATOPS-plus.git
cd TRICERATOPS-plus && pip install .
```

Tkinter. Included with conda Python. On Ubuntu/WSL: `sudo apt install python3-tk`. On Windows 11, WSL GUI windows appear on the desktop (WSLg).

5.2 Starting

```
cd /path/to/your/pipeline/folder
python triceratops_gui.py
```

The GUI reads every option from `build_parser()` in the validation script in the same folder and restores the last session. Optional: `python triceratops_gui.py --script /path/to/triceratops_validation.py`. Check the install with **Tools** → **Run regression tests**.

6. A tour of the window

From top to bottom: menu bar (File, Profiles, Tools, Help); run environment (Python interpreter and working folder); tabs for options (Target & ephemeris—including Mission/KIC for Kepler—Light curve, Shape & tests, Field, TRICERATOPS, Ground photometry); then Run & log and Results. Hover an option name for its flag and help text. The command line is updated live; Copy puts it on the clipboard.

6.1 Profiles

A profile stores settings for one object under `profiles/`. Choosing a profile resets options not listed in it to defaults, so the same profile always yields the same command. Save with File → Save profile (includes interpreter and working folder) or Save target profile (options only).

7. Worked example: AAVSO campaign 942 (TIC 65344135)

Load the supplied profile for TIC 65344135 (or enter TIC, ephemeris P = 36.88 d, T0, depth, T14, enable LD fit and a suitable ephemeris scan width). Set the interpreter to your triceratops environment (or triplus for TRICERATOPS+). Add ground light curves on the Ground photometry tab when available. Run and inspect the log and the Results tab (summary JSON and output files). Example products for this target are included in the repository under `TIC65344135/` (summary, log, key plots).

8. Notes on the TESS-only result for TIC 65344135

With TESS alone the neighbour contribution can leave a non-negligible NFPP. Ground photometry that clears the neighbour drives NFPP toward zero and raises the planet-on-target probability. See the example summary and plots in the repository for the numerical values of a given run date.

9. Other useful functions

- **Import command line** — File → Import command line fills the fields from a pasted command.
- **Copy command** — for a terminal or notes.
- **Predicted transit times** — Tools menu; uses BTJD or BKJD according to Mission.
- **Check interpreter** — reports triceratops / NumPy / pytransit versions.
- **Run regression tests** — pytest on tests/.
- **Audit** — runs triceratops_audit.py on an output folder when available.
- **Manuals** — Help opens this guide and the technical manual if present next to the script.

Part III

Reference

10. All options

Tables list the options of `triceratops_validation.py` **v4.2**, grouped as on the GUI tabs. Empty GUI fields mean "use the default". The GUI reads the live parser, so new flags appear automatically.

Tip. *Options most observers need: TIC (or Mission + KIC), known ephemeris, scan half-width, limb-darkened fit, ground light curves; for TRICERATOPS+ often dilution mode none with aperture depth.*

Tab: Target & ephemeris

GUI field	Command line	Default	Meaning
TIC ID	--tic TIC	—	TIC of the target (optional in Kepler mode if --kic is set).
Mission (TESS / Kepler)	--mission tess|kepler	tess	tess or kepler. Kepler: long-cadence per quarter, BKJD times; requires --kic.
KIC ID (Kepler)	--kic N	—	Kepler Input Catalog number (--mission kepler).
Host star for TRICERATOPS (TIC)	--host-tic TIC	—	Host for TRICERATOPS if not the light-curve target (e.g. close companion).
Stellar parameters from	--star-source tic|kepler	tic	R, M, Teff, logg from TIC or Kepler catalogues.
Manual R M Teff [logg]	--stellar …	—	Manual host parameters (3 or 4 numbers).
Output folder	--outdir DIR	—	Default TIC<tic>/ or KIC<kic>/.
Known ephemeris	--ephemeris P T0 depth_ppm T14_h	—	Skips TLS search. T0: BTJD/BKJD or full BJD_TDB.
Blind TLS search	--blind-search	off	Force TLS search.
TLS period min/max [d]	--p-min / --p-max	—	Narrow the TLS grid.
Ephemeris check	--ephem-search auto|scan|grid|off	auto	How to check/refine P, T0 before detrending.
No ephemeris check	--no-ephem-scan	off	Same as --ephem-search off.
Scan half-width [h]	--ephem-scan-hours H	—	Default max(2×T14, 12 h).
Grid half-range in P [d]	--ephem-grid-dp D	—	Grid search range in period.
Grid min. S/N	--ephem-grid-min-snr X	—	Default 7.
Do not refine P, T0	--no-refine-ephemeris	off	Keep input ephemeris.
Refine: max chi2_red	--refine-chi2-max X	—	Default 8.
Fixed mid-times	--no-tc-fit	off	Do not fit per-transit mid-times.

Tab: Light curve

GUI field	Command line	Default	Meaning
Light-curve source	--lc-source tesscut|sap|pdcsap	tesscut	Kepler mode switches default away from tesscut.
TESScut cache folder	--download-dir DIR	—	Cache for cutouts / SPOC products.
Exclude sectors	--exclude-sectors S …	—	Drop sectors before extraction.
No automatic sector exclusion	--no-auto-exclude-sectors	off	Keep all sectors.
Max sector rms	--sector-max-rms	—	Default 0.02.
Aperture	--aperture optimal|percentile	optimal	TESScut aperture choice.
Dilution correction	--dilution-mode sector|none	sector	sector: put each sector on the target flux scale.
Raw aperture flux	--raw-aperture-flux	off	dilution_corrected=False.
Light curve is dilution corrected	--dilution-corrected	off	Tell TRICERATOPS so.
Detrend windows [d]	--detrend-windows …	—	Multi-window blind TLS.
Error floor [ppm]	--err-floor-ppm	0	Floor on flux uncertainty.

GUI field	Command line	Default	Meaning
Do not inflate sigma	--no-inflate	off	No chi2_red=1 inflation.

Tab: Shape & tests

GUI field	Command line	Default	Meaning
Limb-darkened fit	--ld-fit	off	LD model as reference for sigma inflation.
LD coefficients free	--ld-free	off	Fit q1, q2.
LD coefficients u1 u2	--ld-coeffs U1 U2	—	Fixed quadratic LD.
No secondary-eclipse search	--no-secondary-search	off	Skip secondary search.
Secondary min. S/N	--secondary-min-snr X	—	Default 7.
Red-noise scrambles	--red-noise-scrambles N	0	Phase-randomisation test.
Red-noise seed	--red-noise-seed	42	RNG seed.
No timing analysis	--no-etv	off	Skip ETV / depth variation.
Extra mid-times file	--extra-times FILE	—	Ground mid-times for ETV.

Tab: Field

GUI field	Command line	Default	Meaning
Contrast curve	--contrast-curve FILE	—	sep [″], Δmag.
Vet neighbours	--vet-neighbours	off	Vet from neighbours' own LCs.
No difference imaging	--no-difference-imaging	off	Skip centroids.
Exclude stars (TIC)	--exclude-stars TIC …	—	Cleared elsewhere.

Tab: TRICERATOPS

GUI field	Command line	Default	Meaning
Min. runs (n_iter)	--n-iter	20	Minimum calc_probs runs.
Samples per run (N)	--n-samples N	—	Monte-Carlo draws per scenario.
Do not pool runs	--no-pool	off	Mean of per-run FPPs instead of pooled evidence.
Sensitivity runs	--sensitivity	off	FPP at several sigma scales.
Runs per sensitivity case	--sens-runs	3	Runs per scale.
Model exposure [min]	--tri-exptime-min M	—	Exposure smoothing of models.
Max. runs	--tri-max-iter N	—	Cap while not converged.
Time budget [s]	--tri-time-budget S	—	Default 3600.
TRILEGAL / field-star file	--trilegal-fname FILE	—	Skip online TRILEGAL.
TRICERATOPS+	--tri-plus auto|on|off	auto	Pass ground LCs to calc_probs.

Tab: Ground photometry

GUI field	Command line	Default	Meaning
Ground light curves	--ground-lc SPEC	—	Repeatable; see keys below.
Ground list file	--ground-list FILE	—	One SPEC per line.
Do not remove cleared stars	--ground-no-clear	off	Report only.
Mid-time window [min]	--ground-tc-window	—	Default 60.
Detection S/N	--ground-min-snr	—	Default 5.
Depth tolerance	--ground-depth-tol	—	Default 0.25.
Default seeing [″]	--ground-fwhm	—	Default 3.0.

SPEC keys: path, band, star, ap, fwhm, obs, time, fcol, ecol, tcol, mag, detrend, ra, dec, pixscale, plus, label. Example: `--ground-lc "file.csv,band=R,star=65344134,ap=3.5,fwhm=2.4"`

Kepler quick example

```
python triceratops_validation.py --mission kepler --kic 3755522 \
--ephemeris 10.5 245.3 1200 3.2 --ld-fit
```

11. Output files

All files go to the output folder (default `TIC<tic>/` or `KIC<kic>/` in Kepler mode).

File	Content
triceratops_output_TIC<tic>.txt	Full log ending with SUMMARY.
summary_<tic>.json	Key results (Results tab).
phased_transit_*.pdf, ld_fit_*.pdf	Folded LC and limb-darkened fit.
per_transit_depths_*.pdf/.txt	Per-transit depth and mid-time.
secondary_search_*.pdf/.txt	Secondary-eclipse search.
etv_*.pdf/.txt	Timing and depth variation.
difference_imaging_*.csv / *.pdf	Centroid offsets.
ground_clearance_*.pdf/.csv	Ground clearance per star.
ground_lightcurves_*.pdf	Ground transit fits.
plot_fits_*.pdf	TRICERATOPS scenario fits.
audit_*.pdf/.json/.txt	Optional post-run audit.

12. Troubleshooting

Problem	Solution
GUI: No module named tkinter	conda Python, or sudo apt install python3-tk.
triceratops not importable	Activate the correct environment; Tools → Check interpreter.
pytransit / evaluate_pv errors	pip install "pytransit==2.2.0".
NumPy 2.x issues with TRICERATOPS+	conda install "numpy<2.4" in triplus.
TESScut slow or failing	Set --download-dir; retries are automatic.
Push/repo too large	Do not commit tesscut caches, full LC dumps or FITS; only summary, log, key plots.

13. Glossary

Term	Meaning
FPP	False positive probability (not a planet on the target).
NFPP	Nearby false positive probability (resolved neighbours).
BTJD / BKJD	BJD_TDB – 2457000 (TESS) / – 2454833 (Kepler).
TP / EB / NEB	Transiting planet / eclipsing binary / nearby EB scenarios.
TRICERATOPS+	Fork that accepts external multi-colour light curves in calc_probs.
PASSED	FPP < 0.015 and NFPP < 0.001 (Giacalone et al. 2021).

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Please cite Giacalone et al. (2021) for TRICERATOPS and the pipeline version string from the run log.